D - I like Increasing

分值：700 分

Problem Statement

You are given a permutation $P = (P_1, P_2, \ldots, P_N)$ of $(1, 2, \ldots, N)$.

给定一个长度为 $(1, 2, \ldots, N)$ 的排列 $P = (P_1, P_2, \ldots, P_N)$。

For a sequence of integers $x = (x_1, x_2, \ldots, x_k)$, define the **score** of x as the number of indices i satisfying $x_i < x_{i+1}$.

对于整数序列 $x = (x_1, x_2, \ldots, x_k)$，定义 x 的**得分**为满足 $x_i < x_{i+1}$ 的下标 i 的数量。

Process the following query Q times.

共需要处理 Q 次如下询问。

- You are given positive integers l and r with $1 \le l \le r \le N$. For $X = (P_l, P_{l+1}, \ldots, P_r)$, solve the following problem.
  - Let M be the maximum **score** of a non-empty subsequence of X. Find the minimum length of a non-empty subsequence of X whose **score** equals M.
- 给定正整数 l 和 r，满足 $1 \le l \le r \le N$。对于 $X = (P_l, P_{l+1}, \ldots, P_r)$，求解以下问题：
  - 设 M 是 X 的非空子序列能达到的最大**得分**，求所有得分等于 M 的 X 的非空子序列中，长度的最小值。

What is a subsequence?

什么是子序列?

A subsequence of a sequence A is a sequence obtained by removing zero or more elements from A and arranging the remaining elements in their original order.

序列 A 的子序列是指从 A 中删除零个或多个元素后，剩余元素保持原有顺序排列得到的序列。

Constraints

- $1 \le N, Q \le 2 \times 10^5$

- P is a permutation of $(1, 2, \ldots, N)$.
- $1 \leq l \leq r \leq N$
- All input values are integers.
- $1 \leq N, Q \leq 2 \times 10^5$
- P 是 $(1, 2, \ldots, N)$ 的一个排列。
- $1 \leq l \leq r \leq N$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

N Q
P_1 P_2 $\ldots$ P_N
query_1
query_2
$\vdots$
query_Q

Each query is given in the following format:

每个询问按照以下格式给出：

l r

Output

Output Q lines. The i-th line should contain the answer to query_i.

输出 Q 行，第 i 行应包含第 query_i 个询问的答案。

Sample Input 1

```
6 4
2 1 4 6 3 5
```

```
1 3
3 6
2 2
1 6
```

Sample Output 1

```
2
4
1
5
```

For the first query, $X = (2, 1, 4)$. The maximum **score** $M = 1$ is achieved by $(2, 4), (1, 4), (2, 1, 4)$. The minimum length among these is 2, achieved by $(2, 4), (1, 4)$.

对于第一个询问，$X = (2, 1, 4)$。最大得分 $M = 1$ 可由子序列 $(2, 4), (1, 4), (2, 1, 4)$ 达到，其中长度最小的是 2，由子序列 $(2, 4), (1, 4)$ 达到。

For the second query, $X = (4, 6, 3, 5)$ and $M = 2$. $(4, 6, 3, 5)$ achieves the minimum length 4 among those with **score** 2.

对于第二个询问，$X = (4, 6, 3, 5)$，$M = 2$。在所有得分等于 2 的子序列中，$(4, 6, 3, 5)$ 达到了最小长度 4。

For the third query, $X = (1)$ and $M = 0$. (1) achieves the minimum length 1 with **score** 0.

对于第三个询问，$X = (1)$，$M = 0$。子序列 (1) 达到了最小长度 1，其得分为 0。

For the fourth query, $X = (2, 1, 4, 6, 3, 5)$ and $M = 3$. $(2, 4, 6, 3, 5)$ achieves the minimum length 5 with **score** 3.

对于第四个询问，$X = (2, 1, 4, 6, 3, 5)$，$M = 3$。子序列 $(2, 4, 6, 3, 5)$ 达到了最小长度 5，其得分为 3。

Sample Input 2

```
12 8
8 3 5 7 9 6 11 1 10 4 12 2
3 4
```

```
10 11
5 8
3 8
4 10
2 10
5 7
1 8
```

Sample Output 2

```
2
2
2
4
5
7
2
5
```